

Flexmock: Fast, easy, stockable smocking method using 3D printed self-shrinkable pattern sheet

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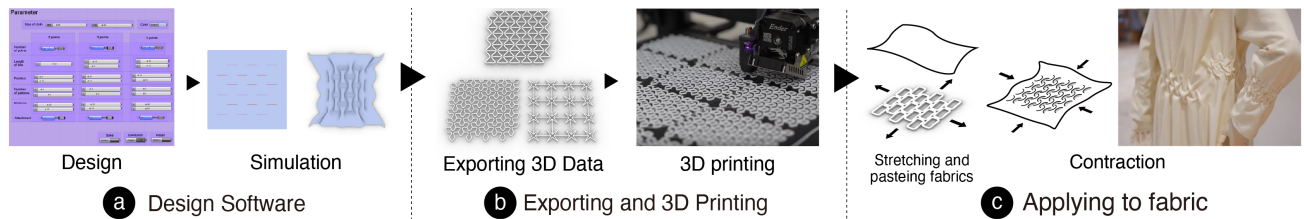


Figure 1: Flexmock workflow. (a) Design of pattern line while checking sewing simulation results. (b) 3D printing of attachment data. (c) Pasting attachment onto fabric and forming a smocking pattern using shrinkage force.

ABSTRACT

In this study, we propose Flexmock, a fabrication method that enables users to easily generate smocking patterns onto cloth by shrinking attachment sheets with geometric patterns. The proposed method offers the following advantages: (1) creating smocking patterns parametrically using design software and simulating the finished product; (2) automatically generating and exporting data for self-shrinking attachments; and (3) quickly applying smocking patterns to fabrics via attached 3D printed pattern sheets.

CCS CONCEPTS

• **Applied computing** → **Computer-aided manufacturing**; • **Human-centered computing** → **Interactive systems and tools**.

KEYWORDS

Smocking, digital fabrication, 3D printing, computational design

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1 INTRODUCTION

First developed in the Middle Ages, smocking is a sewing technique primarily used to decorate clothing. In the traditional hand-stitched smocking process, *pattern lines*, multiple regularly arranged lines, are drawn on the fabric surface to support sewing, and the endpoints of the lines are sewn together, creating three-dimensional shapes on the fabric surface. One challenge associated with smocking production is the difficulty of predicting sewing results. Attempts made to address this issue include a method that predicts results by simulating the sewing process [3] and a computer simulation that generates pattern lines and curved concrete molds using smocking [4]. Another challenge associated with smocking is the complexity of sewing. Research on the deformation of fabric surfaces has enabled the automation of shape changes using steam [6] and heat [5]. Furthermore, Haynes et al. [2] automated the contraction of smocking endpoints using Shape Memory Alloys (SMAs). However, these methods require specialized machines or manual sewing, incurring the challenges of requiring specialized skills and long production times.

2 OVERVIEW

Flexmock is a fabrication method that allows users to quickly and easily apply smocking patterns to fabric by shrinking 3D printed attachment sheets with geometric patterns. The workflow of fabric deformation using Flexmock is shown in Figure 1. First, a design software is used to design the pattern line while checking the sewing simulation results (Figure 1a). Next, the design data are exported and 3D printed to apply smocking to cloth using a stretchable material (Figure 1b). Finally, the attachment in stretched

state is pasted onto the fabric using an adhesive, and the restoring force gives the fabric a three-dimensional shape (Figure 1c).

3 DESIGN SOFTWARE

Using the design software, users can select one of three types (two-, three-, and four-endpoint contractions) of smocking designs and create data for attachment (Figure 1a). The data generation process proceeds as follows: (1) create a pattern line by specifying the position, number of repetitions, and length, and place it on the fabric surface; (2) run a simulation to check the contraction result of the attachment and modify the data; and (3) generate and export modeling data for the attachments. This software was developed using Rhino7 and Grasshopper.

4 ATTACHMENT

An auxetic structure is used as an attachment pattern to contract the endpoints. Figure 2 shows examples of auxetic structures used for the three contraction patterns along with the attachment and application results. Each structure has a negative Poisson's ratio, allowing for attachments that contract and expand in the direction perpendicular to the direction of contraction and expansion (Figure 2c). Stretchable thermoplastic polyurethane (TPU95 1.75 mm, PolyFlex) is used as a filament to print the attachment. General 3D printers (Anycubic Kobra 2 Max, Creality Ender-3 s1 and s1 Plus) are used to produce the attachments, and an adhesive (3M Scotch 6004M) is used for attachment to the cloth.

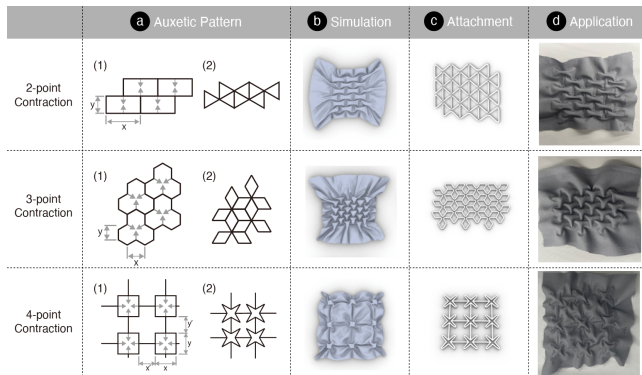


Figure 2: Auxetic pattern diagrams and their applications. (a) Auxetic pattern for each contraction. (b) Smocking simulation results. (c) Attachment data from design software. (d) Application examples to cloth.

5 EVALUATION

To confirm the effects of varying borderline thickness on fabric shrinkage, we created five types of attachment unit patterns with different border thicknesses, where the endpoint of the attachment was deformed and the restoring force was measured using a tensile tester. In this experiment, the restoring force and deformation of the fabric increased along with borderline thickness.

We also applied an attachment sheet to several fabric materials to confirm the reproducibility of smocking with respect to fabric

Fabric	Cotton	Cotton Sheet	Linen	Polyester
Thickness	380 μm	1165 μm	259 μm	136 μm
Fabric	Nylon	Knit	Denim	Leather
Thickness	243 μm	1580 μm	721 μm	624 μm

Figure 3: Application of smocking sheets to multiple clothing materials. The softer and more elastic the fabric, the more distinct the pattern produced.

material. The results shown in Figure 3 demonstrate that smocking can be applied to any material except leather and that the softer and more elastic the material, such as knit, the more beautiful the finish.

The TPU filament used to create attachments in this experiment (Figure 3) shrank within less than 0.5 seconds. The author, who possesses intermediate skill in sewing, required approximately an hour to manually create a pattern with the same effect as the attachment sheet.

6 APPLICATIONS

Flexmock can be used to deform fabrics using the restoring force of the attachment (Figure 4a), as well as providing a means of rapid prototyping by storing smocking patterns on sheets (Figure 4b). We created application examples for fashion items (Figure 4c), cushion covers (Figure 4d), and roll screens (Figure 4e).

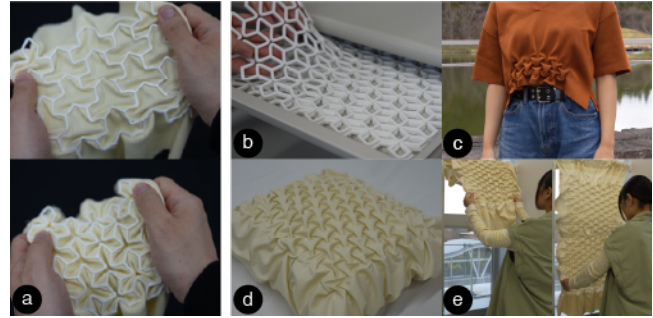


Figure 4: Flexmock application examples. (a) Deformation using restoring force of attachment. (b) Stock of attachments for prototypes. (c) Knit clothes. (d) Cushion cover. (e) Roll screen.

7 FUTURE WORK

The software used in this study enables the design of three attachment patterns (two-, three-, and four-contraction points). In the future, the collection of new smocking patterns and their corresponding auxetic structures into a library [1] can be expected to further expand the applicability of Flexmock.

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